31/01/2025

Code A Phase 1

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot-13, Sector-18, Udyog Vihar, Gurugram, Haryana-122018

MM : 720

NCERT Booster Test Series for NEET (OYM)-2024-25_Test-05

Time : 180 Min.

Topics Covered:
Physics: Work, Energy & Power, System of Particles & Rotational Motion, Current Electricity, Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids, Thermal Properties of Matter
Chemistry: Thermodynamics, Equilibrium, Redox Reactions, The p-Block Elements (Group-13 to 18)
Botany: Biological Classification, Morphology of Flowering Plants, Anatomy of Flowering Plants, Plant Kingdom
Zoology: Body Fluids and Circulation, Excretory Products and Elimination, Locomotion and Movement, Chemical Coordination and Integration

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. The work done by a force on a body does not depend upon

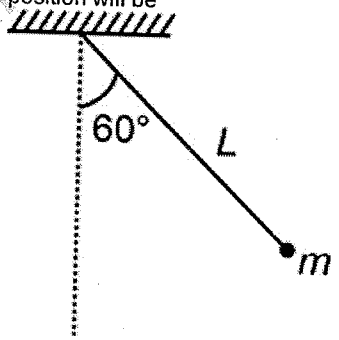
(1) Displacement

(2) Force applied

(3) Angle between force and displacement

(4) Momentum of the body

2. A bob of mass m is suspended by a light string of length L. It is released from the position as shown. The centripetal acceleration of the bob when it reaches bottom-most position will be



(1) $2g$

(2) g

(3) $\frac{g}{2}$

(4) Zero

3. Two identical point masses are moving with velocities $\left(2\hat{i}\right)$ m/s and $\left(2\hat{i}+4\hat{j}\right)$ m/s. The speed of centre of mass of the system is

(1) 2 m/s

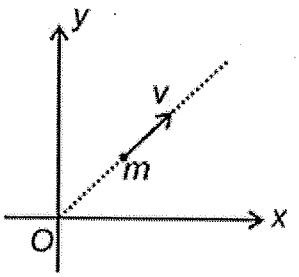
(2) $2\sqrt{2}$ m/s

(3) $4\sqrt{2}$ m/s

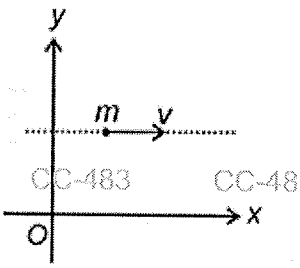
(4) 4 m/s

4. A particle of mass *m* is moving with constant speed *v* along different paths. In which of the following case(s), magnitude of angular momentum of the particle about origin remains constant?

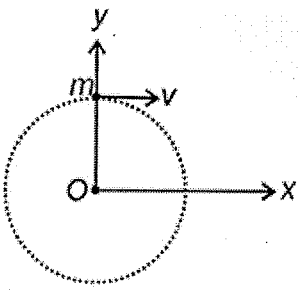
(1)



(2)

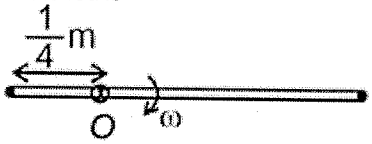


(3)



(4) All of these

5. A uniform rod of length 1 m and mass 2 kg is hinged at point '*O*' as shown. If it rotates with 12 rad/s, then kinetic energy of the rod is



(1) 42 J

(2) 9 J

(3) 81 J

(4) 21 J

6. The correct expression for drift speed of electrons in a conductor is (symbols have their usual meanings)

(1) $\frac{eE^2}{m}\tau$

(2) $\frac{eE}{m^2}\tau$

(3) $\frac{eE}{m}\tau$

(4) $\frac{eE}{m}\tau^2$

7. The storage battery of a car has an emf of 12 V. If the internal resistance of the battery is 0.6 Ω, the maximum current that can be drawn from the battery is

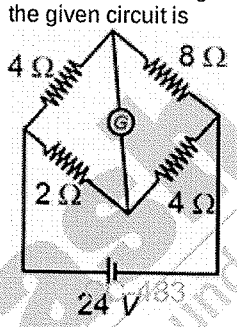
(1) 20 A

(2) 10 A

(3) 6 A

(4) 9 A

8. The current flowing through the 8 Ω resistor as shown in the given circuit is



(1) 4 A

(2) 6 A

(3) 2 A

(4) 5 A

9. A planet is orbiting around sun in an elliptical orbit. If the angular momentum of the planet with respect to sun at perihelion is L_p and at the aphelion is L_a, then

(1) $L_p > L_a$

(2) $L_p < L_a$

(3) $L_p = L_a$

(4) Data insufficient

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10. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): Mars takes more time to complete one revolution of the sun than earth.
Reason (R): Time period of revolution (T) and orbital radius (R) are related as $T \propto R^3$.
In the light of the above statements, choose the most **appropriate** answer from the options given below:

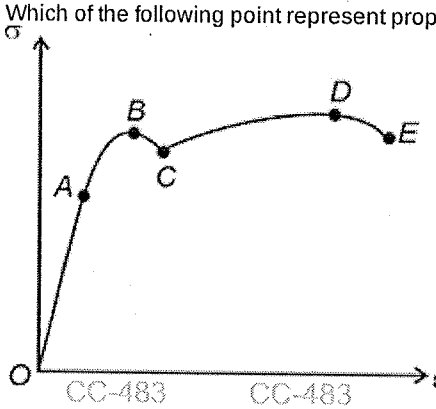
(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct

(4) (A) is not correct but (R) is correct

11. A typical stress-strain curve for a metal is given below. Which of the following point represent proportional limit?



(1) A

(2) B

(3) D

(4) E

12. Consider the following statements:
Statement (A): Bulk modulus is relevant for solids, liquids and gases.
Statement (B) : Shear modulus is only relevant for liquids.
Statement (C): Young's modulus is relevant for solids only.
The **correct** statement(s) is/are

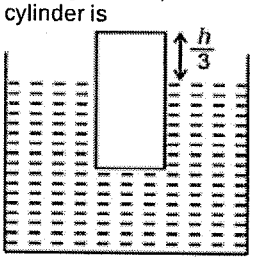
(1) Only (A)

(2) Only (A) and (B)

(3) Only (C)

(4) Only (A) and (C)

13. A cylinder of area of cross-section A_0 and height h is floating on a liquid of density ρ as shown. The mass of the cylinder is



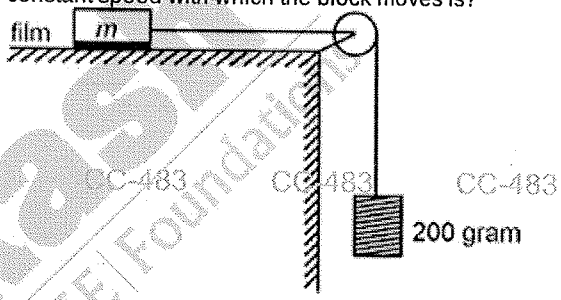
(1) $\frac{\rho h A_0}{3}$

(2) $\frac{3\rho h A_0}{4}$

(3) $\frac{4\rho h A_0}{3}$

(4) $\frac{2\rho h A_0}{3}$

14. A metal block of base area 0.1 m^2 is placed on a table having a liquid film of thickness 0.4 mm between them. If the coefficient of viscosity of liquid is 5×10^{-3} Pa s, then the constant speed with which the block moves is?



(1) 0.8 m s^{-1}

(2) 2.2 m s^{-1}

(3) 1.01 m s^{-1}

(4) 1.6 m s^{-1}

15. The unit of surface tension is

(1) $\frac{N}{m}$

(2) $\frac{J}{m^2}$

(3) kg s^{-2}

(4) All of the above

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16. 100 g ice at 0°C is mixed with 100 g water at 80°C. The final temperature of the mixture is

(1) 40°C

(2) 25°C

(3) 0°C

(4) 47.14°C

17. The coefficient of volume expansion of glycerine is $50 \times 10^{-5}\ K^{-1}$. The fractional change in its density for a 50°C rise in temperature is

(1) 0.25%

(2) 2.5%

(3) 3.5%

(4) 0.35%

18. Negative of work done by conservative forces is equal to change in

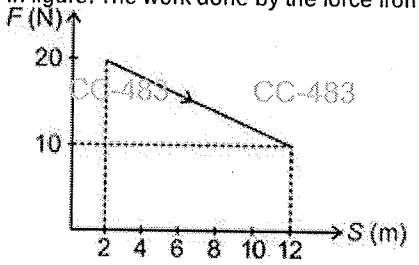
(1) Potential energy

(2) Kinetic energy

(3) Mechanical energy

(4) Both (2) and (3)

19. The force-displacement (*F*-*S*) graph of a body is as shown in figure. The work done by the force from 2 m to 12 m is



(1) 200 J

(2) 150 J

(3) 100 J

(4) 175 J

20. The work done by a force $\vec{F} = \left(-6x^3\hat{i}\right)$ N in displacing a particle from $x = 4$ m to $x = -2$ m is

(1) –240 J

(2) 360 J

(3) 420 J

(4) –360 J

21. The moment of inertia of a circular disc about its diameter is 10 kg m^2, then the moment of inertia of same disc about an axis tangential to disc and perpendicular to its plane will be

(1) 40 kg m^2

(2) 50 kg m^2

(3) 120 kg m^2

(4) 60 kg m^2

22. What is torque of a force $\vec{F} = 7\hat{i} + 3\hat{j} - 5\hat{k}$ about the origin? The force acts on the particle whose position vector is $\left(\hat{i} - \hat{j} + \hat{k}\right)$.

(1) $2\hat{i} + 2\hat{j} + 10\hat{k}$

(2) $2\hat{i} + 12\hat{j} + 4\hat{k}$

(3) $2\hat{i} + 12\hat{j} + 10\hat{k}$

(4) $8\hat{i} + 12\hat{j} + 10\hat{k}$

23. The ratio of orbital speed of two satellites of earth of equal masses and revolving with orbital radii *R* and 4*R* respectively is

(1) 1 : 8

(2) 2 : 1

(3) 1 : 4

(4) 4 : 1

24. If kinetic energy *K* and potential energy *U* is possessed by an orbiting satellite, then

(1) $K < 0$

(2) $U < 0$

(3) $K = 0$

(4) $U = 0$

25. Choose the correct statement.

(1) Gravitational potential as well as gravitational field inside the shell are not zero

(2) Only gravitational field inside a shell is zero

(3) Gravitational field and gravitational potential both are zero inside a shell

(4) Only gravitational potential inside the shell is zero

26. A wire can be broken by applying a load of 300 N. The force required to break another wire of same length and same material but double in diameter, will be

(1) 300 N

(2) 600 N

(3) 900 N

(4) 1200 N

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27. Modulus of elasticity depends on

(1) Nature of material

(2) Shape and size of material

(3) Force applied on the material

(4) Both (2) & (3)

28. Breaking stress of a steel wire of radius 1 mm and length 1 m is 10^{12} N/m^2. Breaking stress of another steel wire of radius 2 mm and length 0.5 m will be

(1) 4×10^{12} N/m^2

(2) 10^{12} N/m^2

(3) 2.5×10^{11} N/m^2

(4) 2×10^{12} N/m^2

29. The excess pressure inside one soap bubble is 3 times than inside a second soap bubble, then the ratio of their surface area is

(1) 1 : 9

(2) 1 : 27

(3) 1 : 3

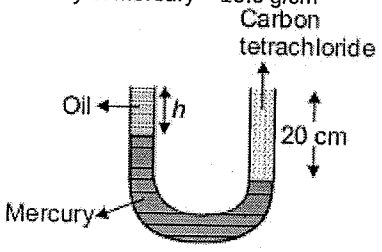
(4) 3 : 1

30. Calculate the value of h in U-tube shown in the following figure.

Density of oil = 0.9 g/cm^3

Density of carbon tetrachloride = 1.6 g/cm^3

Density of mercury = 13.6 g/cm^3



(1) 18.9 cm

(2) 20.9 cm

(3) 16.0 cm

(4) 14.9 cm

31. The temperature of a body changes from 60°C to 40°C in 7 minutes with temperature of surroundings being 10°C. Its temperature after next 7 minutes is (Assuming Newton's law of cooling to be valid)

(1) 28°C

(2) 48°C

(3) 50°C

(4) 40°C

32. If a bimetallic strip is heated, it will

(1) Bend towards the metal with lower thermal expansion coefficient

(2) Bend towards the metal with higher thermal expansion coefficient

(3) Twist itself into helix

(4) Have no bending

33. Ratio of linear expansion coefficient (α), areal expansion coefficient (β) and volume expansion coefficient (γ) for an isotropic material is

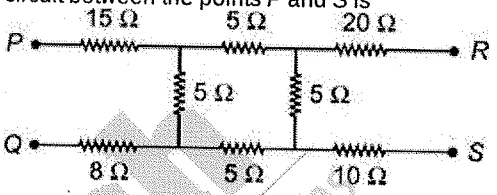
(1) 1 : 3 : 2

(2) 1 : 2 : 3

(3) 4 : 2 : 3

(4) 3 : 2 : 1

34. In the following figure, the equivalent resistance of the circuit between the points P and S is



(1) 55 Ω

(2) 30 Ω

(3) 48 Ω

(4) 34 Ω

35. A conductor having cuboidal shape, has dimensions in ratio 1 : 2 : 5, the ratio of maximum to minimum resistance across two opposite faces is

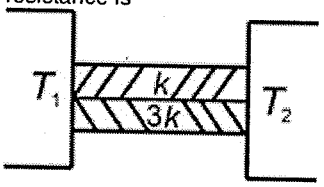
(1) 5 : 1

(2) 25 : 1

(3) 10 : 1

(4) 5 : 2

36. Two conducting rods each of area of cross-section A and length l are connected as shown. If thermal conductivities of the rods are k and $2k$, then the equivalent thermal resistance is



(1) $\frac{4l}{kA}$

(2) $\frac{l}{kA}$

(3) $\frac{3l}{4kA}$

(4) $\frac{l}{4kA}$

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37. In a hydraulic press, the cross-sectional area of the two cylinders is 2×10^{-2} m^2 and $10^{-1} m^2$. A force of 100 N is applied at the small cylinder, the pressure produced at the larger cylinder is?

(1) 2 kPa

(2) 5 kPa

(3) 1 kPa

(4) 2.5 kPa

38. A steel wire of length L is stretched by length l. The energy stored in the steel wire per unit volume will be (Young's modulus of steel wire is Y)

(1) $\frac{1}{2}\frac{Yl^2}{L^2}$

(2) $\frac{Yl^2}{L}$

(3) $\frac{1}{2}\frac{Yl^2}{L}$

(4) $\frac{1}{2}\frac{Y^2l}{L}$

39. Acceleration due to gravity on the surface of earth is g. In column-I different values of acceleration due to gravity is given and in column-II different values of height and depth are given.
Match the columns and tick the **correct** option.
(R is radius of earth)

	Column-I		Column-II
a	$\frac{g}{2}$	(i)	Depth = R
b.	$\frac{g}{4}$	(ii)	Depth = $\frac{R}{2}$
c.	$\frac{g}{9}$	(iii)	Height = $2R$
d.	Zero	(iv)	Height = R

(1) a(ii), b(iii), c(iv), d(i)

(2) a(ii), b(iv), c(iii), d(i)

(3) a(i), b(ii), c-(iv), d(iii)

(4) a(iv), b(iii), c(i), d(ii)

40. Two bulbs are connected in series with a battery and the power dissipation through them is in the ratio of 1 : 2, the potential drop across them is in the ratio of

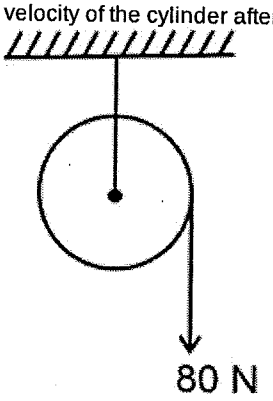
(1) 1 : 2

(2) 1 : 1

(3) 2 : 3

(4) 4 : 1

41. A string of negligible mass is wound on the rim of a solid cylinder of mass 10 kg and radius 20 cm. The string is pulled with a constant force of 80 N as shown. Angular velocity of the cylinder after 2 seconds will be



(1) 80 rad/s

(2) 90 rad/s

(3) 160 rad/s

(4) 120 rad/s

42. Three particles each of 1 kg mass are moving with acceleration 1 m/s^2, 2 m/s^2 and 3 m/s^2 in the same direction. The net force acting on this three-particle system is

(1) 9 N

(2) 12 N

(3) 6 N

(4) 3 N

43. Consider the following statements:
Statement (A): The total linear momentum is conserved at each instant of any collision.
Statement (B): The kinetic energy of the system is conserved at each instant of any collision.
Choose the **correct** option.

(1) Only statement (A) is correct

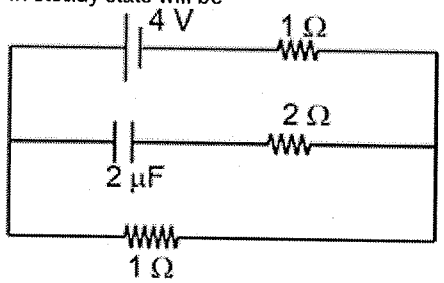
(2) Only statement (B) is correct

(3) Both statements are correct.

(4) Both statements are incorrect

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44. The energy stored in 2 μF capacitor as shown in the circuit in steady state will be



(1) 4 μJ

(2) 2 μJ

(3) 0.5 μJ

(4) 1 μJ

45. Work done by frictional force

(1) Is always negative

(2) Is always positive

(3) Maybe positive, negative or zero

(4) Is zero

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CHEMISTRY

46. Match the column I of the given compounds with column II containing the names of the processes involved in their manufacture

	Column I		Column II
a.	Chlorine gas	(i)	Ostwald's process
b.	Ammonia gas	(ii)	Contact process
c.	Sulphuric acid	(iii)	Deacon's process
d.	Nitric acid	(iv)	Haber's process

Choose the correct match

(1) a(iii), b(iv), c(i), d(ii)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(iv), b(iii), c(i), d(ii)

(4) a(iv), b(i), c(ii), d(iii)

47. Total number of six- membered rings and five-membered rings present in fullerene (C-60) respectively are

(1) 12 and 20

(2) 16 and 20

(3) 20 and 15

(4) 20 and 12

48. The reaction for which the relation $\Delta H \neq \Delta U$ is true, is

(1) $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

(2) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

(3) $C(s) + O_2(g) \rightarrow CO_2(g)$

(4) $N_2(g) + O_2(g) \rightarrow 2NO(g)$

49. Oxidation state of the two chlorine atoms in $CaOCl_2$ are

(1) +1 & –1

(2) –1 & –1

(3) +1 & +1

(4) +3 & –1

50. Consider the following reaction :

$C(\text{graphite}) + CO_2(g) \rightarrow 2CO(g)$

ΔH and ΔS for the given reaction is 170 kJ and 170 JK^{-1} respectively. The reaction will be spontaneous at

(1) 520 K

(2) 720 K

(3) 820 K

(4) 1020 K

51. Consider the following statements:
Statement I : For free expansion of an ideal gas in vacuum under adiabatic condition, q = 0 and $\Delta U \neq 0$
Statement II : For free expansion of an ideal gas in vacuum; w = 0
In the light of above statements choose the correct option

(1) Both statement I and statement II are correct

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are incorrect

52. Which of the following are state functions?
a. H – TS
b. q
c. w
d q + w

(1) a & b only

(2) b & c only

(3) a, b & c only

(4) a & d only

53. Choose the correct statement:

(1) If $\Delta H > 0$ and $\Delta S < 0$, then reaction will be spontaneous at all temperatures

(2) In any spontaneous process, entropy of universe is always decrease

(3) Endothermic reactions are always non-spontaneous

(4) The entropy of any pure crystalline substance approaches zero as the temperature approaches absolute zero

54. Consider the given reactions and their respective enthalpies
$2CO(g) + O_2(g) \rightarrow 2CO_2(g)$ $\Delta H° = -x$ kJ
$C(s) + O_2(g) \rightarrow CO_2(g)$ $\Delta H° = -y$ kJ
The enthalpy of formation of CO (in kJ) is

(1) x – 2y

(2) $\frac{2y-x}{2}$

(3) $\frac{x-2y}{2}$

(4) y – 2x

55. The pair of compounds which can make basic buffer is

(1) NH_4OH & NH_4Cl

(2) CH_3COOH & CH_3COONa

(3) H_2CO_3 & Na_2CO_3

(4) $HClO_4$ & $NaClO_4$

56. Consider the following statement:
Statement I : Density is an intensive property
Statement II : Pressure and volume both are extensive properties
In the light of above statements choose the correct option

(1) Both statement I and statement II are correct

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are incorrect

57. pH of 0.1 M aqueous solution of potassium formate is (pK_a of formic acid = 3.75)

(1) 9.75

(2) 11.2

(3) 4.75

(4) 8.37

58. Consider the following statement:
Statement I : SiO_2 and GeO_2 are acidic oxides
Statement II : PbO_2 and CO are neutral oxides.
In the light of above statements choose the correct option

(1) Both statement I and statement II are correct

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are incorrect

59. If equal volumes of two solutions of pH = 2 and pH = 3 are mixed, then the pH of resulting solution will be (log 5.5. = 0.74)

(1) 0.74

(2) 2.74

(3) 3.26

(4) 2.26

60. For dissociation of $NH_4HS(s)$ into $H_2S(g)$ and $NH_3(g)$,

$$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$$

The total pressure at equilibrium is 20 atm. The K_c for this equilibrium will be

(1) $10(RT)^2$

(2) $100(RT)^{-2}$

(3) 10RT

(4) 100 RT

61. Consider the given equilibrium

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ + heat

Which of the given conditions will favour the formation of $SO_3(g)$?

(i) Decrease in temperature
(ii) Increase in pressure
(iii) Removal of oxygen
(iv) Addition of SO_3

Choose the appropriate answer

(1) (i) & (iii) only

(2) (i) & (ii) only

(3) (i), (ii) & (iii) only

(4) (i), (ii), (iii) & (iv)

62. Which of the given salts will undergo cationic hydrolysis?

(1) Ammonium sulphate

(2) Sodium acetate

(3) Sodium chloride

(4) Calcium acetate

63. Consider the following process

Process	ΔH (kJ/mol)
A → B	x
B + C → D	–y
E + 2C → 2D	z

For; A + B → E, ΔH will be

(1) x + 2y – z

(2) x – 2y + z

(3) –x – y – z

(4) x – 2y – z

64. If K_a of monobasic weak acid HA is 1.75×10^{-5}, then the hydrolysis constant of 0.01 M NaA solution is

(1) 1.75×10^{-11}

(2) 5.7×10^{-11}

(3) 1.75×10^{-10}

(4) 5.7×10^{-10}

65. Heat of combustion of solid benzoic acid at constant volume is –x kJ at 100°C. The heat of combustion at constant pressure is

(1) (x – 2RT) kJ

(2) $\left(\frac{3}{2}RT + x\right)$ kJ

(3) $\left(\frac{5}{2}RT - x\right)$ kJ

(4) (x + 2RT)kJ

66. Given below are the two statements

Statement I: In P_4 molecule the bond angles are 60°.

Statement II: White phosphorus is non-poisonous and insoluble in carbon disulphide.

In the light of the above statements, choose the correct answer.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is incorrect but statement II is correct

(4) Statement I is correct but statement II is incorrect

67. Chlorine on reaction with cold and dilute NaOH gives

(1) $NaClO_3$ + NaCl

(2) NaOCl + $NaClO_3$

(3) NaCl + $NaClO_4$

(4) NaCl + NaOCl

68. Partial hydrolysis of XeF_6 produces

(1) XeO_3

(2) XeF_4

(3) $XeOF_4$

(4) XeO_4

69. Which among the following has highest boiling point?

(1) H_2O

(2) CH_4

(3) HF

(4) NH_3

70. Which of the following does not contain N-N bond?

(1) N_2O

(2) N_2O_3

(3) N_2O_4

(4) N_2O_5

71. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)** :

Assertion (A) : Boron cannot form BF_6^{3-} ion.

Reason (R) : Boron does not have empty d-orbital in outermost shell

In the light of above statements choose the most appropriate answer from options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true, (R) is false

(4) (A) is false, (R) is true

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72. Select the correct order as per property mentioned.

(1)	Si > Ge > Sn > C	:	Melting point
(2)	C-C > Si-Si > Sn-Sn > Ge-Ge	:	Bond energy
(3)	Sn > Si > Ge > C	:	Covalent radius
(4)	C >> Si > Ge ≈Sn	:	Catenation power

(1) (1)

(2) (2)

(3) (3)

(4) (4)

73. For the redox reaction,

$xFe^{2+} + y\,Cr_2O_7^{2-} + zH^+ \rightarrow Fe^{3+} + Cr^{3+} + H_2O$,

x, y and z are respectively

(1) 3, 1, 4

(2) 6, 1, 7

(3) 6, 2, 14

(4) 6, 1, 14

74. Incorrect statement about graphite is

(1) It has layered structure

(2) Each carbon atom is sp^3 hybridised

(3) It conducts electricity

(4) It is very soft and slippery

75. The disproportionation reaction among the following is

(1) $2F_2 + 2OH^- \rightarrow 2F^- + OF_2 + H_2O$

(2) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

(3) $Cl_2 + OH^- \rightarrow ClO^- + Cl^- + H_2O$

(4) $Fe + 3H_2O \rightarrow Fe_2O_3 + 3H_2$

76. If at equilibrium HI dissociates 50%, then K_{eq} for the reaction $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$ is

(1) 4

(2) $\frac{1}{2}$

(3) $\frac{1}{4}$

(4) $\frac{1}{8}$

77. Match List-I with List-II.?

	List-I (Salt)		List-II (Ksp)
(a)	AgBr	(i)	$27S^4$
(b)	Hg_2Cl_2	(ii)	$4S^3$
(c)	$Al(OH)_3$	(iii)	S^2
(d)	$Ca_3(PO_4)_2$	(iv)	$108S^5$

Choose the **correct** answer form the options given below

(1) a(iv), b(ii), c(iii), d(i)

(2) a(ii), b(iii), c(i), d(iv)

(3) a(iii), b(ii), c(i), d(iv)

(4) a(i), b(iii), c(iv), d(ii)

78. For the equilibrium,

$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$

If the value of the equilibrium constant (K_c) is 4×10^{-6} at 1100 K then K_p for the reaction at the same temperature will be (Given R = 0.082 L atm K^{-1} mol^{-1})

(1) 4.8×10^{-3}

(2) 3.6×10^{-4}

(3) 7.5×10^{-2}

(4) 1.2×10^{-1}

79. Combination redox reaction among the following is

(1) $2KClO_3 \rightarrow 2KCl + 3O_2$

(2) $3Mg + N_2 \rightarrow Mg_3N_2$

(3) $Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr$

(4) $Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2$

80. The percent ionisation of 0.01 M acetic acid in 0.1M HCl will be (K_a of acetic acid is 1.8×10^{-5})

(1) 0.18%

(2) 0.009%

(3) 0.018%

(4) 0.09%

81. E°_{cell} of the cell, $Ni(s)\,|\,Ni^{2+}(aq)\,||\,Au^{3+}(aq)\,|\,Au(s)$ will be

(Given: $E^\circ_{Ni/Ni^{2+}} = +0.25\ V$ and $E^\circ_{Au^{3+}/Au} = 1.4\ V$)

(1) 1.65 V

(2) 1.15 V

(3) –1.65 V

(4) –1.15 V

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82. Example of homogeneous equilibrium is

(1) $Ca(OH)_2(s) \rightleftharpoons Ca^{2+}(aq) + 2OH^-(aq)$

(2) $CH_3COOC_2H_5(aq) + H_2O(l) \rightleftharpoons CH_3COOH(aq) + C_2H_5OH(aq)$

(3) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

(4) $H_2O(l) \rightleftharpoons H_2O(g)$

83. Which of the given compounds has highest acidic strength?

(1) HOCl

(2) $HClO_2$

(3) $HClO_3$

(4) $HClO_4$

84. Choose the incorrect match:

(1) $Zn(OH)_2$ – White

(2) $Fe_2O_3 . xH_2O$– Brown

(3) $[Cu(NH_3)_4]^{2+}$ – Deep blue

(4) $[Ag(NH_3)_2]Cl$ – Yellow

85. All of the given species have equal number of lone pair of electrons on central atom, except

(1) $XeOF_4$

(2) XeF_4

(3) XeO_3

(4) XeF_6

86. Match the List I of the given reactions with their respective gaseous product given in List II

	List-I		List-II
a.	$NH_4NO_3 \xrightarrow{\Delta}$	(i)	NO_2
b.	$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$	(ii)	NH_3
c.	$(NH_4)_2SO_4 + NaOH \rightarrow$	(iii)	N_2O
d.	$I_2 + HNO_3 \xrightarrow{\Delta}$	(iv)	N_2

Choose the correct answer from the options given below

(1) a(iv), b(i), c(ii), d(iii)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(i), c(iv), d(ii)

(4) a(iii), b(iv), c(ii), d(i)

87. In which of the following case change in entropy is negative?

(1) $Na(s) \rightarrow Na(g)$

(2) Expansion of gas at constant temperature

(3) Boiling of water

(4) $2Cl(g) \rightarrow Cl_2(g)$

88. A gas is allowed to expand in a well insulated container against a constant external pressure of 1 atm from an initial volume 2L to a final volume of 4L. The change in internal energy ΔU of the gas in joule will be

(1) –100 J

(2) +100 J

(3) –202 J

(4) +202 J

89. The species which does not act as a Lewis acid is

(1) BF_3

(2) $AlCl_3$

(3) H_3BO_3

(4) NH_3

90. Choose the incorrect statement about phosphine gas.

(1) It has rotten fish like odour

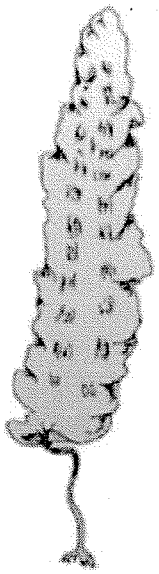
(2) It is highly poisnous

(3) It is blue colour gas

(4) It is used in Holme's signals

BOTANY

91. Choose the **correct** ones w.r.t. the given plant.



(a) Many species of it are used as food.
(b) It majorly possesses chlorophyll *a*, *d* and fucoxanthin
(c) Cell wall of the given plant is made of cellulose and algin.
(d) Gametes are pyriform and bear two laterally attached flagella
(e) The commercial product obtained from the given plant is used to grow microbes and in preparations of jellies

(1) (a), (b) and (c) only
(2) (b), (d) and (e) only
(3) (c) and (d) only
(4) (a), (c) and (d) only

92. Select the **correctly** matched pair.

(1)	*Cycas*	–	Male and female cones are borne on different trees
(2)	*Selaginella*	–	Microphyllous leaves and heterosporous
(3)	*Funaria*	–	A liverwort, used as fuel and as a packing material
(4)	*Ulothrix*	–	Colonial alga and isogamous

(1) (1)
(2) (2)
(3) (3)
(4) (4)

93. Protonema and prothallus differ from each other as the latter is

(1) An independent, photosynthetic, gametophytic stage
(2) Developed after spore germination in the terrestrial plants that have vascular tissue but do not produce seeds.
(3) A conspicuous large, multicellular thalloid stage which bears male and female sex organs
(4) A predominant stage of the life cycle of a moss

94. Choose the **incorrect** statement about the infectious agent, discovered by T.O. Diener.

(1) It is a causative agent of potato spindle tuber disease
(2) It is similar in size to viruses
(3) It lacks capsid that is found in viruses
(4) Its a free RNA of low molecular weight

95. Among the following organisms, an example of flagellated protozoan is

(1) *Paramoecium*
(2) *Plasmodium*
(3) *Trypanosoma*
(4) *Entamoeba*

96. Read the given statements.
(a) Subsidiary cells are the specialised chloroplast containing epidermal cells present in the vicinity of guard cells.
(b) Barrel shaped cells with intercellular spaces have depositions of suberin in root.
(c) Vascular bundles form the conducting tissue and translocate water, minerals and food material
(d) In stem of grasses, the peripheral vascular bundles are generally larger than the centrally located ones.
(e) Pith is large and well developed in roots of all flowering plants.
How many of the above statements are **incorrect**?

(1) One
(2) Two
(3) Four
(4) Three

97. Stamens are attached to the perianth in the flowers of

(1) *Petunia*
(2) *Indigofera*
(3) Lily
(4) Brinjal

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98. In arid regions, a fleshy cylindrical modified stem, which carries out photosynthesis can be found in

(1) Grapevines

(2) *Opuntia*

(3) *Colocasia*

(4) *Euphorbia*

99. Match **List I** with **List II**

	List-I		List-II
a.	Cacti	(i)	Thorns protect plant from browsing animals
b.	*Bougainvillea*	(ii)	Leaves are small and short-lived
c.	*Rhizophora*	(iii)	Spines are modified leaves
d.	Australian acacia	(iv)	Roots come out of the ground and grow vertically upwards

Choose the **correct** option

(1) a(ii), b(i), c(iii), d(iv)

(2) a(i), b(iii), c(iv), d(ii)

(3) a(iii), b(ii), c(iv), d(i)

(4) a(iii), b(i), c(iv), d(ii)

100. Read the given assertion (A) and reason (R) and choose the **correct** option.
Assertion (A) : The roots found in wheat plant are known as fibrous roots.
Reason (R) : In wheat, the primary root is short lived and is replaced by a large number of roots originating from base of the stem.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

101. Cell walls form two thin overlapping shells, which fit together as in a soap box in

(1) Slime moulds

(2) Diatoms

(3) Dinoflagellates

(4) Euglenoids

102. Natural system of classification was

(1) Based on evolutionary relationships between various organisms

(2) Given by George Bentham and Joseph Dalton Hooker

(3) Based on a few morphological characters only

(4) Given by Linnaeus

103. Small specialised roots called coralloid roots that are associated with N_2- fixing cyanobacteria are found in

(1) *Cedrus*

(2) *Ginkgo*

(3) *Cycas*

(4) *Pinus*

104. In seeds of maize, a structure named scutellum is a/an

(1) Sheath which encloses plumule

(2) Large, shield shaped cotyledon

(3) Outer covering of endosperm separating the embryo

(4) Scar on the seed coat, through which developing seeds are attached to the fruit

105. The flowers in which the margin of thalamus grows upward enclosing the ovary completely, and the other parts of the flower arise above the ovary are

(1) Plum, Rose, Peach

(2) Guava, Cucumber, Ray florets of sunflower

(3) Mustard, Peach, China rose

(4) Brinjal, Guava, Rose

106. The meristem which occurs at the tip of a shoot

(1) Is a cylindrical meristem and produces woody axis

(2) Regenerates parts that are removed by the grazing herbivores

(3) Is also responsible for forming axillary bud

(4) Is responsible for producing the secondary tissues

107. Select the **correctly** matched pair.

(1) Epiblema – Many of its cells protrude in the form of multicellular trichomes in stem

(2) Conjoint vascular bundles – Phloem located on the outer side of xylem

(3) Bast fibres – Elongated, branched and have pointed apices

(4) Guard cells – Bean shaped in monocots

(1) (1)

(2) (2)

(3) (3)

(4) (4)

108. Vessels and tracheids share a common characteristic feature, that is both
(a) Are found in all angiosperms and gymnosperms
(b) Have thick lignified walls and are devoid of protoplasm
(c) Are the main water transporting elements in flowering plants
(d) Are interconnected through perforations in their common walls
Choose the option with the **correct** one(s)

(1) (b) only

(2) (a), (b) and (c) only

(3) (b) and (c) only

(4) All except (b)

109. The bundle sheath which surrounds the vascular bundles is

(1) Found in both dorsiventral leaf and stem of monocots

(2) Thin-walled in leaf and constituted by sclerenchymatous cells

(3) Identical in structure to mesophyll cells of a shoot system

(4) Absent in monocotyledonous plants

110. Select the **odd** one w.r.t. companion cells.

(1) These cells help in maintaining the pressure gradient in the sieve tubes

(2) These cells are specialized parenchymatous cells

(3) These cells are found in gymnosperms, pteridophytes and angiosperms

(4) The functions of sieve tubes are controlled by the nucleus of companion cells

111. Select the **incorrect** one about the plant group, in which the ovules are not enclosed by any ovary wall and remain exposed, both before and after fertilisation.

(1) Male and female gametophytes do not have an independent free-living existence.

(2) *Sequoia* is one of the tallest tree species

(3) Thick cuticle and sunken stomata help to reduce water loss

(4) The plants in this group are heterosporous as well as homosporous

112. Sclereids are

(1) Commonly found in the leaves of tea

(2) Simple tissues that form major components within the plant organs

(3) Thin-walled and made up of cellulose

(4) Absent in the pulp of fruits like guava

113. Starch sheath is found in the

(1) Hypodermis of monocot stem

(2) Innermost layer of the cortex in dicot stem

(3) Endodermis of dicot root

(4) Pericycle of dicot stem

114. Select the **odd** one out w.r.t ground tissue system.

(1) Pith

(2) Medullary rays

(3) Vascular bundles

(4) Pericycle

115. All of the below given options exemplify the tissue which leads to increase in the girth of plant, **except**

(1) Fascicular vascular cambium

(2) Interfascicular cambium

(3) Intercalary meristem

(4) Cork cambium

116. Match List I with List II

	List-I		List-II
a.	*Pinus*	(i)	Elaborate mechanism of spore dispersal, present in capsule
b.	*Polysiphonia*	(ii)	The female gametophytes are retained on the parent sporophytes for variable periods
c.	*Funaria*	(iii)	Male and female gametophytes do not have an independent free-living existence
d.	*Salvinia*	(iv)	Complex post fertilisation developments and lacks embryo

Choose the **correct** option

(1) a(ii), b(i), c(iii), d(iv)

(2) a(iii), b(ii), c(iv), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(ii), b(i), c(iv), d(iii)

117. Arrange the following zones in a sequential manner according to their position from outer to inner side of transverse section of a stem. Choose the **correct** option.
(a) The cells of the layer are rich in starch grains.
(b) The layer is covered with a thin cuticle bearing few stomata
(c) The cells of the layer are sclerenchymatous in the form of semi-lunar patches.
(d) A large number of rounded, parenchymatous cells with large intercellular spaces occupies the central portion.
(e) Few layers of living cells which are much thickened at the corners.

(1) d → c → a → b → e

(2) e → b → d → a → c

(3) b → e → a → c → d

(4) b → e → c → a → d

118. The element of phloem, which is absent in most of the monocotyledons is

(1) Sieve tubes

(2) Companion cell

(3) Xylem fibres

(4) Phloem parenchyma

119. In which of the given sets of plants, underground stems are modified to store food in them?

(1) *Colocasia*, turnip, garlic

(2) Sweet potato, turmeric, potato

(3) Ginger, zaminkand, *Colocasia*

(4) Carrot, potato, onion

120. The medicinal plant, called muliathi belongs to the family, which shows

(1) Berry or capsule type of fruit

(2) Diadelphous androecium

(3) Actinomorphic bisexual flowers

(4) Bicarpellary superior ovary

121. Root hairs arise from the zone, that

(1) Is proximal to the region of elongation with respect to base of the stem

(2) Is covered by the thimble-like structure

(3) Undergoes rapid elongation and enlargement

(4) Is few millimetres above the root cap

122. In mango, the fruit is

(1) One seeded but have dry pericarp

(2) Formed without fertilisation of ovary

(3) Arised from monocarpellary superior ovary

(4) Having fibrous mesocarp like coconut

123. In mustard,

(a) There may be variation in length of filaments within a flower
(b) Formation of false septum in the ovary occurs
(c) Tap root system is seen
(d) A pair of leaves arise at each node and lie opposite to each other
(e) Flowers can be divided into two similar halves only in one particular plane
Choose the **correct** ones

(1) (b), (d) and (e) only

(2) (c) and (d) only

(3) All except (d) and (e)

(4) (a) and (b) only

124. Formation of specialized structures called gemmae occurs in

(1) *Sphagnum*

(2) *Marchantia*

(3) *Polytrichum*

(4) *Funaria*

125. Both *Gelidium* and *Gracilaria*,

(1) Reproduce asexually by non-motile spores

(2) Lack phycoerythrin pigment in their body

(3) Store food in the form of mannitol

(4) Belong to a class phaeophyceae

126. Which of the following is the smallest living cell known and can survive without oxygen?

(1) Mycoplasma

(2) *Nostoc*

(3) *Chlorella*

(4) Cyanobacteria

127. The kingdom whose boundaries are not well defined and also their members reproduce asexually and sexually by a process involving cell fusion and zygote formation includes

(1) Unicellular eukaryotes

(2) All plants

(3) Archaebacteria and blue-green algae

(4) Fungi and eubacteria

128. In the five kingdom classification system of Whittaker, there is no mention of which of the given organisms?
(i) Lichens
(ii) Viruses
(iii) Prions
(iv) Viroids
Choose the **correct** option.

(1) All (i), (ii), (iii), (iv)

(2) (ii) only

(3) (ii), (iii) only

(4) (i), (iv) only

129. Rust and smut fungi respectively are

(1) *Claviceps* and *Trichoderma*

(2) *Puccinia* and *Ustilago*

(3) *Alternaria* and *Rhizopus*

(4) *Colletotrichum* and *Puccinia*

130. Read the following statements:
Statement A : Corolla are usually brightly coloured to attract insects for pollination and to protect the flower in the bud stage.
Statement B : A lateral branch with short internodes and each node bearing a rosette of leaves and a tuft of roots is found in *Pistia* and *Eichhornia*
In the light of above statements, choose the **correct** option.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

131. Choose the **correct** option w.r.t plants with their respective placentation and aestivation.

(1) *Asparagus*-parietal, twisted

(2) Tobacco - axile, valvate

(3) Pea-marginal, twisted

(4) Belladonna-parietal, vexillary

132. Select the **mismatched** pair.

(1) *Albugo* – Parasitic fungi, causes white spots on mustard leaves

(2) Bladderwort – Partially heterotrophic plant

(3) Protozoans – Heterotrophs and live as predators or parasites

(4) *Neurospora* – Causes sleeping sickness

(1) (1)

(2) (2)

(3) (3)

(4) (4)

133. Read the following statements:
Statement A : The outermost layer of stele is made up of sclerenchyma and parenchyma respectively in roots and stems of dicotyledonous plants.
Statement B : Collenchyma provides mechanical support to the growing parts of the plant such as the young stem and petiole of a leaf.
In the light of above statements, choose the **correct** option.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

134. Read the following features of the member(s) of a particular class of fungi and choose the **correct** option for it.
(a) Edible and are considered delicacies
(b) Mycelium is branched and septate
(c) Occurrence of dikaryotic stage
(d) Conidia produced exogenously on the special mycelium
These features are most likely to be exhibited by

(1) *Trichoderma*

(2) Morels and truffles

(3) *Saccharomyces*

(4) Puffballs

135. A protist named *Euglena* does **not**

(1) Have a protein rich layer

(2) Behave like heterotroph *i.e* by predating on other smaller organisms

(3) Act as the chief 'producers' in the oceans

(4) Have two unequal flagella.

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ZOOLOGY

136. In human blood, the total percentage of plasma proteins is equal to the percentage of _____ in total WBCs. Select the option that correctly fills the blank.

(1) Monocytes

(2) Lymphocytes

(3) Neutrophils

(4) Eosinophils

137. Thrombin is formed from an inactive substance 'X'. Conversion of 'X' is catalysed by an enzyme complex named

(1) Prothrombin

(2) Thrombokinase

(3) Fibrin

(4) Fibrinogen

138. In human heart, atrium and ventricle of same side are separated by the

(1) Thick fibrous tissue

(2) Thin fibrous and muscular tissue

(3) Thin muscular and avascular wall

(4) Thick, non-muscular and non-fibrous tissue

139. During a normal cardiac cycle in humans, just before the beginning of atrial contraction, ventricles are

(1) Nearly two third filled by blood

(2) Only one-third filled by blood

(3) Completely filled by blood

(4) Empty

140. Blood pressure is lowest in which of the following blood vessels?

(1) Systemic aorta

(2) Renal artery

(3) Pulmonary artery

(4) Vena cava

141. Which of the following statements is incorrect w.r.t. cardiovascular system of humans?

(1) Heart sounds and ECG have a clinical significance.

(2) If the heart rate is 90 times per minute, then the duration of each cardiac cycle will be 0.8 sec.

(3) In a normal healthy individual, the systolic blood pressure is 120 mm Hg.

(4) By counting the number of QRS complexes that occur in a given time period, one can determine the heart beat rate of an individual.

142. All of the following are increased in an athlete as compared to a non-athlete under resting conditions, **except**

(1) Stroke volume

(2) Heart rate

(3) Duration of cardiac cycle

(4) Strength of ventricular contraction

143. Each organised skeletal muscle in our body is made of a number of muscle bundles held together by a common collagenous connective tissue layer, called

(1) Fascicles

(2) Fascia

(3) Muscle fibre

(4) Myofibril

144. All of the following are flat bones and form the part of axial skeleton in human skeletal system, **except**

(1) Occipital bone

(2) Rib

(3) Sternum

(4) Patella

145. Match Column I with Column II and choose the correct option.

Column I	Column II
a. 'I' band	(i) Elastic fibre
b. 'A' band	(ii) Central part of thick filament
c. 'H' zone	(iii) Both actin and myosin filaments
d. 'Z' line	(iv) Only actin filament

(1) a(iv), b(iii), c(i), d(ii)

(2) a(iv), b(ii), c(iii), d(i)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(iv), b(i), c(ii), d(iii)

146. White muscle fibres differ from red muscle fibres as former

(1) Posssess plenty of mitochondria

(2) Perform aerobic respiration only

(3) Have high amount of sarcoplasmic reticulum

(4) Contain high amount of red coloured oxygen storing pigment

147. Which of the following is incorrect w.r.t maximally contracted skeletal muscle?

(1) Length of A-band remains unchanged

(2) Reduction in the length of myosin filament

(3) Reduction in the length of 'I' band

(4) Disappearance of 'H' zone

148. Read the following statements carefully w.r.t. myosin filament.

A. It is made up of many monomeric proteins named meromyosins.

B. It provides binding site for ATP and actin.

C. It is thicker as compared to the actin filament

D. Heavy meromyosin is made up of only globular head and a short tail

Choose the option with incorrect statement(s) only.

(1) Only D

(2) A and B

(3) B and C

(4) Only C

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149. Choose the correct option to complete the analogy.
Tubular secretion : Selective process : : Glomerular filtration : _____

(1) Non-selective process

(2) Selective process

(3) Active transport

(4) Osmosis

150. Select the incorrect match w.r.t human excretory system.

(1) Counter current mechanism – Henle's loop and vasa recta

(2) Descending limb of loop of Henle – Permeable to water

(3) DCT – Osmolarity of filtrate may be 200 $mOsmolL^{-1}$

(4) Bowman's capsule – Contains modified muscle cells called podocytes

(1) (1)

(2) (2)

(3) (3)

(4) (4)

151. In humans, urea is produced from ammonia through metabolic reactions within

(1) Stomach

(2) Liver

(3) Pancreas

(4) Kidney

152. In ureotelic animals such as humans

(1) Urea is completely excreted out from kidney matrix

(2) Urea cannot be retained in the kidney matrix

(3) Some amount of urea may be retained in the kidney matrix

(4) Urea is excreted in the form of pellet or paste with a minimum loss of water

153. Maximum number of action potentials is generated by _____ in human heart.
Choose the correct option to fill in the blank.

(1) SAN

(2) AVN

(3) Purkinje fibres

(4) Bundle of His

154. How many organisms mentioned in the box below has/have protonephridia as the excretory organ?

Planaria, *Amphioxus*, Prawn, Cockroach

Choose the correct option.

(1) One

(2) Two

(3) Three

(4) Four

155. Read the following statements w.r.t human excretory system.
Statement A : Right kidney is lower as compared to the left kidney.
Statement B : Kidneys are situated between the levels of last thoracic and third lumbar vertebrae close to the ventral inner wall of the abdominal cavity.
Choose the correct option.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

156. Cortical nephron can be differentiated from juxtamedullary nephron in all of the following aspects, **except**

(1) Former forms the majority of nephron in humans than latter

(2) Latter has longer Henle's loop than former

(3) As compared to former, latter possesses Henle's loop that runs parallel to vasa recta deep into the medulla

(4) Former plays a major role in producing highly concentrated urine than latter

157. Which of the following is a common symptom between individuals suffering from diabetes mellitus and diabetes insipidus?

(1) Hyperglycemia

(2) Glycosuria

(3) Uremia

(4) Diuresis

158. The ascending limb of loop of Henle is _____ to water but allows transport of electrolytes _____.
Choose the correct option to fill in the blanks respectively.

(1) Permeable, only actively

(2) Impermeable, only passively

(3) Impermeable, actively or passively

(4) Permeable, actively or passively

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159. How many of the hormones mentioned in the box below affect menstrual cycle in humans?

FSH, Thyroxine, Melatonin, Progesterone, LH

Select the correct option.

(1) Three

(2) Two

(3) Four

(4) Five

160. Choose the source gland of the hormone which acts on the smooth muscles of uterus to stimulate the vigorous contractions during the process of child birth.

(1) Hypothalamus

(2) Pars distalis

(3) Pars intermedia

(4) Ovary

161. After kidney transplantation, which of the following hormones could be administered to the patient to increase the success rate of kidney transplantation and prevent graft rejection?

(1) Mineralocorticoid

(2) Cortisol

(3) ADH

(4) ANF

162. Complete the analogy by choosing the correct option.

Steroid hormone : Progesterone : : Amino-acid derivative : _____

(1) Epinephrine

(2) Growth hormone

(3) TSH

(4) Erythropoietin

163. Intracellular receptors are present for which of the following sets of hormones?

(1) Insulin, glucagon

(2) Adrenaline, noradrenaline

(3) Cortisol, insulin

(4) Testosterone, thyroxine

164. Which pair of hormones generally regulate cellular metabolism by generating second messengers?

(1) Melatonin, somatostatin

(2) Thyroxine, cortisol

(3) Estrogen, thymosin

(4) Aldosterone, progesterone

165. A cranial bone named 'X' has depression in which endocrine gland named 'Y' is present. Identify 'X' and 'Y' respectively and select the correct option.

(1) Temporal, hypothalamus

(2) Ethmoid, thyroid

(3) Sphenoid, pituitary

(4) Sphenoid, pineal

166. The incorrect pair of antagonistic hormone is

(1) Adrenaline and noradrenaline w.r.t. blood sugar level

(2) Parathyroid hormone and calcitonin w.r.t. blood calcium level

(3) Insulin and glucagon w.r.t. blood glucose level

(4) Gastrin and GIP w.r.t. gastric motility

167. Underproduction of hormones from which structure causes acute weakness and fatigue leading to Addison's disease?

(1) Adrenal cortex

(2) Pineal gland

(3) Thyroid gland

(4) Pituitary gland

168. Both the lobes of thyroid gland are interconnected with a

(1) Thin flap of epithelial tissue only

(2) Thin flap of connective tissue

(3) Thick flap of epithelial tissue only

(4) Thick flap of neural tissue

169. Consider the following features:

(a) A type of granulocyte

(b) Possess bilobed nucleus

(c) Constitute 2-3% of the total WBCs and associated with allergic reactions

All of the above features are true for

(1) Eosinophils

(2) Basophils

(3) Neutrophils

(4) Monocytes

170. How many of the features listed below is/are true w.r.t girdles in humans?

a. Pectoral girdle has two bones

b. Pectoral girdle is made up of two separated halves and each half is made up of 2 bones

c. Pelvic girdle consists of two coxal bones and each of the coxal bone is formed by fusion of the ilium, pubis and ischium.

Select the correct option.

(1) Zero

(2) One

(3) Two

(4) Three

171. If the cardiac output of a person is 6.3 L/min, end diastolic volume of left ventricle is 160 mL and end systolic volume of left ventricle is 90 mL , then what is the heart rate of this person?

(1) 80 beats/min

(2) 90 beats/min

(3) 100 beats/min

(4) 70 beats/min

172. Read the following statements w.r.t human blood.
Statement A : All individuals have antibodies in their plasma against the antigen which is present on the surface of their RBCs.
Statement B : All individuals have antibodies in their plasma against Rh-antigen since birth.
Choose the correct option.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

173. What would be the immediate effect, if we cut the chordae tendineae of mitral valve?

(1) First heart sound 'lub' will not be heard

(2) The second heart sound 'dub' will not be heard as it is associated with closure of mitral valve

(3) The flow of blood would decrease in aorta

(4) The flow of blood would increase in pulmonary artery

174. Choose the incorrect option w.r.t the total number of respective bones in an adult human.

(1) Metatarsals + Tarsals = 12

(2) Facial bones + Cervical vertebrae = 21

(3) Vertebrochondral ribs + Floating ribs = 10

(4) Unpaired bones of cranium = 4

175. Match the following columns w.r.t disorders of circulatory system.

	Column I		Column II
a.	Coronary Artery Disease	(i)	Heart stops beating
b.	Angina	(ii)	Congestion of lungs
c.	Heart failure	(iii)	Acute chest pain
d.	Cardiac arrest	(iv)	Atherosclerosis

Select the correct option.

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(iii), c(i), d(ii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(iii), b(ii), c(i), d(iv)

176. When neural signal reaches the neuro-muscular junction, it causes release of 'X' into sarcoplasm which binds to 'Y' and remove the masking of active sites on actin for myosin. Identify 'X' and 'Y' and choose the correct option.

	'X'	'Y'
(1)	Ca^{+2}	Tropomyosin
(2)	Acetylcholine	Subunit of troponin
(3)	Ca^{+2}	Subunit of troponin
(4)	Acetylcholine	Tropomyosin

(1) (1)

(2) (2)

(3) (3)

(4) (4)

177. During muscle contraction, cross-bridge is broken between actin and myosin when

(1) Ca^{+2} binds to tropomyosin

(2) ATP binds with the head of myosin

(3) There is hydrolysis of ATP into ADP and Pi

(4) ATP binds with the actin filament

178. Consider the following
(i) Possesses fluid-filled synovial cavity
(ii) Allows considerable movement
(iii) Present between carpal and metacarpal of human thumb
All of the above features are true for

(1) Saddle joint

(2) Pivot joint

(3) Hinge joint

(4) Gliding joint

179. **Assertion (A) :** Bone has very hard matrix and cartilage has slightly pliable matrix.
Reason (R) : The matrix of bone possesses calcium salts while that of cartilage has chondroitin salts.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

180. Which of the following is a part of nephron but is not considered as a part of renal tubule?

(1) Bowman's capsule

(2) PCT

(3) Collecting duct

(4) Glomerulus

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Code-A

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NCERT BOOSTER TEST SERIES

(for NEET-2025)

Test – 5

Answers

1. (4)	37. (2)	73. (4)	109. (1)	145. (3)
2. (2)	38. (1)	74. (2)	110. (3)	146. (3)
3. (2)	39. (2)	75. (3)	111. (4)	147. (2)
4. (4)	40. (1)	76. (3)	112. (1)	148. (1)
5. (4)	41. (3)	77. (3)	113. (2)	149. (1)
6. (3)	42. (3)	78. (2)	114. (3)	150. (4)
7. (1)	43. (1)	79. (2)	115. (3)	151. (2)
8. (3)	44. (1)	80. (3)	116. (3)	152. (3)
9. (3)	45. (3)	81. (1)	117. (3)	153. (1)
10. (3)	46. (2)	82. (2)	118. (4)	154. (2)
11. (1)	47. (4)	83. (4)	119. (3)	155. (3)
12. (4)	48. (2)	84. (4)	120. (2)	156. (4)
13. (4)	49. (1)	85. (2)	121. (1)	157. (4)
14. (4)	50. (4)	86. (4)	122. (3)	158. (3)
15. (4)	51. (3)	87. (4)	123. (3)	159. (4)
16. (3)	52. (4)	88. (3)	124. (2)	160. (1)
17. (2)	53. (4)	89. (4)	125. (1)	161. (2)
18. (1)	54. (3)	90. (3)	126. (1)	162. (1)
19. (2)	55. (1)	91. (4)	127. (1)	163. (4)
20. (2)	56. (2)	92. (2)	128. (1)	164. (1)
21. (4)	57. (4)	93. (2)	129. (2)	165. (3)
22. (3)	58. (2)	94. (2)	130. (2)	166. (1)
23. (2)	59. (4)	95. (3)	131. (2)	167. (1)
24. (2)	60. (2)	96. (3)	132. (4)	168. (2)
25. (2)	61. (2)	97. (3)	133. (2)	169. (1)
26. (4)	62. (1)	98. (4)	134. (2)	170. (3)
27. (1)	63. (4)	99. (4)	135. (3)	171. (2)
28. (2)	64. (4)	100. (1)	136. (1)	172. (2)
29. (1)	65. (3)	101. (2)	137. (2)	173. (3)
30. (1)	66. (4)	102. (2)	138. (1)	174. (1)
31. (1)	67. (4)	103. (3)	139. (1)	175. (3)
32. (1)	68. (3)	104. (2)	140. (4)	176. (3)
33. (2)	69. (1)	105. (2)	141. (2)	177. (2)
34. (2)	70. (4)	106. (3)	142. (2)	178. (1)
35. (2)	71. (1)	107. (2)	143. (2)	179. (1)
36. (4)	72. (4)	108. (3)	144. (4)	180. (4)

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